

Experiment-8

Aim

To implement an Echo Client-Server communication using TCP and UDP sockets in Java, where the client sends a message and the server echoes it back.

Procedure

Start program execution

Create server:

- TCP → `ServerSocket` with port
- UDP → `DatagramSocket` with port

Wait for client:

- TCP → `accept()`
- UDP → `receive()`

Create client:

- TCP → `Socket` with server IP & port
- UDP → `DatagramSocket` + packet

Client sends message

Server receives it

Server echoes back (TCP: output stream, UDP: `DatagramPacket`)

Client receives and displays response

Repeat if needed

Close sockets and end program

```
T3client.java
File Edit View

import java.net.*;
import java.io.*;
import java.util.*;
public class T3Client
{
    public static void main(String[] args) {
        try{
            Socket Client =new Socket("localhost",5000);
            System.out.println("Client is connected");
        }catch (Exception e){
            System.out.println(e);
        }
    }
}
```

```
T3server.java
File Edit View

import java.net.*;
import java.io.*;
import java.util.*;
class T3Server {
    public static void main(String[] args) {
        try
        {
            ServerSocket ss =new ServerSocket(5000);
            System.out.println("waiting for Client");
            Socket server = ss.accept();
            System.out.println("Server is connected");
        } catch (Exception e) {
            System.out.println(e);
        }
    }
}
```

```
Command Prompt
Microsoft Windows [Version 10.0.22631.5039]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Tcs>cd desktop

C:\Users\Tcs\Desktop>javac T3Client.java

C:\Users\Tcs\Desktop>java T3Client
Client is connected

C:\Users\Tcs\Desktop>|
```

Observation

- TCP establishes a connection before communication
- Ensures reliable, ordered delivery
- UDP requires no connection setup
- Faster, but delivery is not guaranteed
- Both programs successfully echo client messages

Result

Thus, The Echo Client-Server programs using both **TCP and UDP sockets** were successfully implemented and executed. The server correctly received messages from the client and sent back the same message as output.